

Marks Average Demonstration

Title: Marks Average Calculation

Objective/Aim:

To write a Java program to accept marks of five subjects from the user, calculate the total marks, and compute the average marks using the Scanner class.

System Requirements :

Hardware

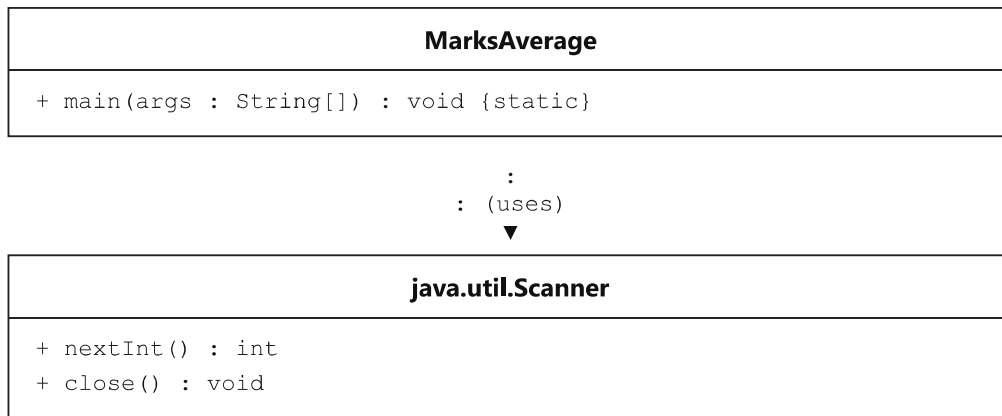
- Standard PC/Laptop
- Intel Core i3/i5 processor or equivalent
- Minimum 4 GB RAM
- Minimum 500 MB free storage
- Keyboard and Monitor

Software

- Operating System: Windows/Linux/macOS
- Java Development Kit (JDK 17 or JDK 21)
- Text Editor or IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code
- Command Prompt/Terminal

Algorithm / Logic Description:

1. Import the Scanner class from `java.util` package to read user input.
2. Create a class named `MarksAverage`.
3. Define the `main()` method.
4. Create a Scanner object named `sc` connected to standard input (`System.in`).
5. Prompt the user and read marks for five subjects (`m1`, `m2`, `m3`, `m4`, `m5`) using `nextInt()`.
6. Add the five subject marks to calculate the total marks: `total = m1 + m2 + m3 + m4 + m5`.
7. Calculate the average marks: `average = total / 5.0`.
8. Display the total marks and the calculated average marks.
9. Close the Scanner object.
10. Stop the program.

UML class Diagram

Relationship: The MarksAverage class depends on and uses the Scanner class to take input values from the console.

Source Code:

```

import java.util.Scanner;

public class MarksAverage {
    public static void main(String[] args) {
        // Creating Scanner object to take user input
        Scanner sc = new Scanner(System.in);

        // Reading marks for 5 subjects
        System.out.print("Enter marks of Subject 1: ");
        int m1 = sc.nextInt();

        System.out.print("Enter marks of Subject 2: ");
        int m2 = sc.nextInt();

        System.out.print("Enter marks of Subject 3: ");
        int m3 = sc.nextInt();

        System.out.print("Enter marks of Subject 4: ");
        int m4 = sc.nextInt();

        System.out.print("Enter marks of Subject 5: ");
        int m5 = sc.nextInt();

        // Calculating total and average
        int total = m1 + m2 + m3 + m4 + m5;
        double average = total / 5.0;

        // Displaying results
        System.out.println("Total Marks = " + total);
        System.out.println("Average Marks = " + average);

        // Closing the scanner
        sc.close();
    }
}
  
```

```
}  
}
```

Compilation & Execution Commands:

Save file as: MarksAverage.java

Compilation: javac MarksAverage.java

Execution: java MarksAverage

Output Screenshot:

```
PS C:\Users\mmeen\RAVI> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' ... 'MarksAverage'  
Enter marks of Subject 1: 70  
Enter marks of Subject 2: 80  
Enter marks of Subject 3: 90  
Enter marks of Subject 4: 95  
Enter marks of Subject 5: 35  
Total Marks = 370  
Average Marks = 74.0  
PS C:\Users\mmeen\RAVI>
```

Result:

The Marks Average program was successfully implemented in Java. The program demonstrates reading user input using the Scanner class, performing arithmetic additions and floating-point divisions, and printing the calculated total and average marks.